

# ALGORITMA & STRUKTUR DATA

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## Praktikum 1

### 2.1

```
1  import java.util.Scanner;
2
3  public class NilaiAkhir {
4
5      Run | Debug
6      public static void main(String[] args) {
7          Scanner sc = new Scanner(System.in);
8
9          System.out.println(x: "Program Menghitung Nilai Akhir");
10         System.out.println(x: "=====");
11
12         System.out.print(s: "Masukkan Nilai Tugas : ");
13         double tugas = sc.nextDouble();
14
15         System.out.print(s: "Masukkan Nilai Kuis : ");
16         double kuis = sc.nextDouble();
17
18         System.out.print(s: "Masukkan Nilai UTS : ");
19         double uts = sc.nextDouble();
20
21         System.out.print(s: "Masukkan Nilai UAS : ");
22         double uas = sc.nextDouble();
23
24         System.out.println(x: "=====");
25
26         // Validasi input
27         if (tugas < 0 || tugas > 100 ||
28             kuis < 0 || kuis > 100 ||
29             uts < 0 || uts > 100 ||
30             uas < 0 || uas > 100) {
31
32             System.out.println(x: "Nilai tidak valid");
33             System.out.println(x: "=====");
34             return; // Resource leak: 'sc' is not closed at this location
35         }
36
37         // ✅ Bobot sesuai contoh (hasil 74)
38         double nilaiAkhir = (0.1 * tugas) +
39                             (0.2 * kuis) +
40                             (0.3 * uts) +
41                             (0.4 * uas);
42
43         // ✅ Pembulatan 1 angka desimal
44         nilaiAkhir = Math.round(nilaiAkhir * 10) / 10.0;
45
46         String huruf;
47
48         if (nilaiAkhir >= 80) huruf = "A";
49         else if (nilaiAkhir >= 73) huruf = "B+";
50         else if (nilaiAkhir >= 65) huruf = "B";
51         else if (nilaiAkhir >= 60) huruf = "C+";
52         else if (nilaiAkhir >= 50) huruf = "C";
53         else if (nilaiAkhir >= 40) huruf = "D";
54         else huruf = "E";
55
56         System.out.println(x: "=====");
57         System.out.printf(format: "Nilai akhir : %.1f\n", nilaiAkhir);
58         System.out.println("Nilai Huruf : " + huruf);
59         System.out.println(x: "=====");
60
61         if (huruf.equals(anObject: "A") || huruf.equals(anObject: "B+") || huruf.equals(anObject: "B") ||
62             huruf.equals(anObject: "C+") || huruf.equals(anObject: "C")) {
```

```

62         System.out.println(x: "SELAMAT ANDA LULUS");
63     } else {
64         System.out.println(x: "ANDA TIDAK LULUS");
65     }
66
67     sc.close();
68 }
69 }

```

```

Program Menghitung Nilai Akhir
=====
Masukkan Nilai Tugas : 85
Masukkan Nilai Kuis : 90
Masukkan Nilai UTS : 120
Masukkan Nilai UAS : 70
=====
Nilai tidak valid
=====
PS G:\Semester 2\ASD>

```

```

Program Menghitung Nilai Akhir
=====
Masukkan Nilai Tugas : 90
Masukkan Nilai Kuis : 40
Masukkan Nilai UTS : 75
Masukkan Nilai UAS : 85
=====
Nilai akhir : 73,5
Nilai Huruf : B+
=====
SELAMAT ANDA LULUS
PS G:\Semester 2\ASD>

```

## 2.2

```

1  import java.util.Scanner;
2
3  public class DeretBilanganNIM {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7
8          System.out.print(s: "Masukkan NIM : ");
9          String nim = sc.nextLine();
10
11          int n = Integer.parseInt(nim.substring(nim.length() - 2));
12
13          if (n < 10) {
14              n += 10;
15          }
16
17          System.out.println("n = " + n);
18          System.out.print(s: "Output : ");
19
20          for (int i = 1; i <= n; i++) {
21              if (i == 10 || i == 15) {
22                  continue;
23              }
24
25              if (i % 3 == 0) {
26                  System.out.print(s: "# ");
27              }
28              else if (i % 2 == 0) {
29                  System.out.print(i + " ");
30              }
31              else {
32                  System.out.print(s: "* ");
33              }
34          }
35
36          sc.close();
37      }
38  }
39

```

```
Masukkan NIM : 2541720102
n = 12
Output : * 2 # 4 * # * 8 # * #
PS G:\Semester 2\ASD> ^C
```

```
Masukkan NIM : 2541720120
n = 20
Output : * 2 # 4 * # * 8 # * # * 14 16 * # * 20
```

2.3

```
1  import java.util.Scanner;
2
3  public class IPSemesterTabel {
4
5      public static String huruf(double n) {
6          if (n > 80) return "A";
7          else if (n > 73) return "B+";
8          else if (n > 65) return "B";
9          else if (n > 60) return "C+";
10         else if (n > 50) return "C";
11         else if (n > 39) return "D";
12         else return "E";
13     }
14
15     public static double bobot(String h) {
16         switch (h) {
17             case "A": return 4.0;
18             case "B+": return 3.5;
19             case "B": return 3.0;
20             case "C+": return 2.5;
21             case "C": return 2.0;
22             case "D": return 1.0;
23             default: return 0.0;
24         }
25     }
26
27     public static void main(String[] args) {
28
29         Scanner sc = new Scanner(System.in);
30
31         String[] mk = {
```

```

32         "Pancasila",
33         "Konsep Teknologi Informasi",
34         "Critical Thinking dan Problem Solving",
35         "Matematika Dasar",
36         "Bahasa Inggris",
37         "Dasar Pemrograman",
38         "Praktikum Dasar Pemrograman",
39         "Keselamatan dan Kesehatan Kerja"
40     };
41
42     int[] sks = {2, 2, 2, 3, 2, 3, 1, 2};
43
44     double[] nilai = new double[mk.length];
45     String[] huruf = new String[mk.length];
46     double[] bobot = new double[mk.length];
47
48     System.out.println(x: "Program Menghitung IP Semester");
49     System.out.println(x: "=====");
50
51     for (int i = 0; i < mk.length; i++) {
52         System.out.print("masukkan nilai Angka untuk MK " + mk[i] + ": ");
53         nilai[i] = sc.nextDouble();
54
55         if (nilai[i] < 0 || nilai[i] > 100) {
56             System.out.println(x: "nilai tidak valid!");
57             return;
58         }
59
60         huruf[i] = huruf(nilai[i]);
61         bobot[i] = bobot(huruf[i]);

```

```

62     }
63
64     System.out.println(x: "\n=====");
65     System.out.println(x: "hasil Konversi Nilai");
66     System.out.println(x: "=====");
67
68     System.out.printf(format: "%-40s %-12s %-12s %-12s\n",
69         ...args: "MK", "Nilai Angka", "Nilai Huruf", "Bobot Nilai");
70
71     double total = 0;
72     int totalSKS = 0;
73
74     for (int i = 0; i < mk.length; i++) {
75         System.out.printf(format: "%-40s %-12.2f %-12s %-12.2f\n",
76             mk[i], nilai[i], huruf[i], bobot[i]);
77
78         total += bobot[i] * sks[i];
79         totalSKS += sks[i];
80     }
81
82     double ips = total / totalSKS;
83
84     System.out.println(x: "=====");
85     System.out.printf(format: "IP : %.2f\n", ips);
86
87     sc.close();
88 }
89 }
90

```

```

Program Menghitung IP Semester
=====
masukkan nilai Angka untuk MK Pancasila: 75
masukkan nilai Angka untuk MK Konsep Teknologi Informasi: 85
masukkan nilai Angka untuk MK Critical Thinking dan Problem Solving: 70
masukkan nilai Angka untuk MK Matematika Dasar: 85
masukkan nilai Angka untuk MK Bahasa Inggris: 85
masukkan nilai Angka untuk MK Bahasa Inggris: 85
masukkan nilai Angka untuk MK Dasar Pemrograman: 62
masukkan nilai Angka untuk MK Praktikum Dasar Pemrograman: 62
masukkan nilai Angka untuk MK Keselamatan dan Kesehatan Kerja: 85

=====
hasil Konversi Nilai
=====

```

| MK                                    | Nilai Angka | Nilai Huruf | Bobot Nilai |
|---------------------------------------|-------------|-------------|-------------|
| Pancasila                             | 75,00       | B+          | 3,50        |
| Konsep Teknologi Informasi            | 85,00       | A           | 4,00        |
| Critical Thinking dan Problem Solving | 70,00       | B           | 3,00        |
| Matematika Dasar                      | 85,00       | A           | 4,00        |
| Bahasa Inggris                        | 85,00       | A           | 4,00        |
| Dasar Pemrograman                     | 62,00       | C+          | 2,50        |
| Praktikum Dasar Pemrograman           | 62,00       | C+          | 2,50        |
| Keselamatan dan Kesehatan Kerja       | 85,00       | A           | 4,00        |

```

=====
IP : 3,47

```

2.4

```

1  public class RoyalGarden {
2
3  public static int hitungPendapatan(int[] stok, int[] harga) {
4      int total = 0;
5      for (int i = 0; i < stok.length; i++) {
6          total += stok[i] * harga[i];
7      }
8      return total;
9  }
10
11 public static void main(String[] args) {
12
13     String[] cabang = {
14         "RoyalGarden 1",
15         "RoyalGarden 2",
16         "RoyalGarden 3",
17         "RoyalGarden 4"
18     };
19
20     String[] bunga = {"Aglonema", "Keladi", "Alocasia", "Mawar"};
21
22     int[][] stok = {
23         {10, 5, 15, 7},
24         {6, 11, 9, 12},
25         {2, 10, 10, 5},
26         {5, 7, 12, 9}
27     };
28
29     int[] harga = {75000, 50000, 60000, 10000};
30
31     System.out.println("=== Pendapatan RoyalGarden ===");

```

```

32     System.out.println(x: "=====");
33
34     for (int i = 0; i < cabang.length; i++) {
35
36         int pendapatan = hitungPendapatan(stok[i], harga);
37
38         String status;
39         if (pendapatan > 150000) {
40             status = "Sangat Baik";
41         } else {
42             status = "Perlu Evaluasi";
43         }
44
45         System.out.printf(format: "%-15s : Rp %,d  -> %s\n",
46             cabang[i], pendapatan, status);
47     }
48 }
49 }
50

```

```

=== Pendapatan RoyalGarden ===
=====
RoyalGarden 1 : Rp 1.970.000 -> Sangat Baik
RoyalGarden 2 : Rp 1.660.000 -> Sangat Baik
RoyalGarden 3 : Rp 1.300.000 -> Sangat Baik
RoyalGarden 4 : Rp 1.535.000 -> Sangat Baik
PS G:\Semester 2\ASD>

```

## TUGAS

1.

```
1  import java.util.Scanner;
2
3  public class Tugas1Plat {
4
5      Run | Debug
6      public static void main(String[] args) {
7          Scanner sc = new Scanner(System.in);
8
9          // Array kode plat
10         char[] KODE = {'A','B','D','E','F','G','H','L','N','T'};
11
12         // Array nama kota (2D)
13         String[] KOTA = {
14             "BANTEN",
15             "JAKARTA",
16             "BANDUNG",
17             "CIREBON",
18             "BOGOR",
19             "PEKALONGAN",
20             "SEMARANG",
21             "SURABAYA",
22             "MALANG",
23             "TEGAL"
24         };
25
26         System.out.print(s: "Masukkan kode plat : ");
27         char input = sc.next().toUpperCase().charAt(index: 0);
28
29         boolean ditemukan = false;
30
31         for (int i = 0; i < KODE.length; i++) {
32             if (input == KODE[i]) {
33                 System.out.println("Kota : " + KOTA[i]);
34                 ditemukan = true;
35                 break;
36             }
37         }
38
39         if (!ditemukan) {
40             System.out.println(x: "Kode tidak ditemukan!");
41         }
42
43         sc.close();
44     }
45 }
```

```
Masukkan kode plat : L
Kota : SURABAYA
```

2.

```
1  import java.util.Scanner;
2
3  public class Tugas2jadwal {
4
5      static Scanner sc = new Scanner(System.in);
6
7      static void inputJadwal(String[][] jadwal, int n) {
8          for (int i = 0; i < n; i++) {
9              System.out.println("\nJadwal ke-" + (i + 1));
10
11              System.out.print(s: "Mata Kuliah : ");
12              jadwal[i][0] = sc.nextLine();
13
14              System.out.print(s: "Ruang      : ");
15              jadwal[i][1] = sc.nextLine();
16
17              System.out.print(s: "Hari        : ");
18              jadwal[i][2] = sc.nextLine();
19
20              System.out.print(s: "Jam          : ");
21              jadwal[i][3] = sc.nextLine();
22          }
23      }
24
25      static void tampilSemua(String[][] jadwal, int n) {
26          System.out.println(x: "\n=== SEMUA JADWAL ===");
27          System.out.printf(format: "%-20s %-15s %-10s %-12s\n",
28              ...args: "Mata Kuliah", "Ruang", "Hari", "Jam");
29
30          for (int i = 0; i < n; i++) {
31              System.out.printf(format: "%-20s %-15s %-10s %-12s\n",
32                  jadwal[i][0], jadwal[i][1], jadwal[i][2], jadwal[i][3]);
33
34          }
35
36      static void cariHari(String[][] jadwal, int n) {
37          System.out.print(s: "\nCari jadwal hari : ");
38          String hari = sc.nextLine();
39
40          boolean ketemu = false;
41
42          for (int i = 0; i < n; i++) {
43              if (jadwal[i][2].equalsIgnoreCase(hari)) {
44                  System.out.println(jadwal[i][0] + " | " + jadwal[i][1] + " | " + jadwal[i][3]);
45                  ketemu = true;
46              }
47          }
48
49          if (!ketemu)
50              System.out.println(x: "Tidak ada jadwal di hari tersebut.");
51      }
52
53      static void cariMatkul(String[][] jadwal, int n) {
54          System.out.print(s: "\nCari mata kuliah : ");
55          String matkul = sc.nextLine();
56
57          boolean ketemu = false;
58
59          for (int i = 0; i < n; i++) {
60              if (jadwal[i][0].equalsIgnoreCase(matkul)) {
61                  System.out.println(jadwal[i][2] + " | " + jadwal[i][1] + " | " + jadwal[i][3]);
62                  ketemu = true;
63              }
64          }
65      }
66  }
```



```

66         if (!ketemu)
67             System.out.println(x: "Mata kuliah tidak ditemukan.");
68     }
69
70     Run | Debug
71     public static void main(String[] args) {
72
73         System.out.print(s: "Jumlah jadwal : ");
74         int n = sc.nextInt();
75         sc.nextLine();
76
77         String[][] jadwal = new String[n][4];
78
79         inputJadwal(jadwal, n);
80
81         int pilih;
82
83         do {
84             System.out.println(x: "\n===== MENU =====");
85             System.out.println(x: "1. Tampilkan semua");
86             System.out.println(x: "2. Cari berdasarkan hari");
87             System.out.println(x: "3. Cari berdasarkan mata kuliah");
88             System.out.println(x: "0. Keluar");
89             System.out.print(s: "Pilih : ");
90
91             pilih = sc.nextInt();
92             sc.nextLine();
93
94             switch (pilih) {
95                 case 1 -> tampilkanSemua(jadwal, n);
96                 case 2 -> cariHari(jadwal, n);
97                 case 3 -> cariMatkul(jadwal, n);
98             }
99
100         } while (pilih != 0);
101
102         sc.close();
103     }
104 }

```

Jumlah jadwal : 2

Jadwal ke-1  
Mata Kuliah : asd  
Ruang : rt7  
Hari : senin  
Jam : 07.30-10.00

Jadwal ke-2  
Mata Kuliah : psi  
Ruang : lpy3  
Hari : selasa  
Jam : 07.00-09.30

===== MENU =====  
1. Tampilkan semua  
2. Cari berdasarkan hari  
3. Cari berdasarkan mata kuliah  
0. Keluar  
Pilih : 1

=== SEMUA JADWAL ===

| Mata Kuliah | Ruang | Hari   | Jam         |
|-------------|-------|--------|-------------|
| asd         | rt7   | senin  | 07.30-10.00 |
| psi         | lpy3  | selasa | 07.00-09.30 |

===== MENU =====  
1. Tampilkan semua  
2. Cari berdasarkan hari  
3. Cari berdasarkan mata kuliah  
0. Keluar  
Pilih : 2

Cari jadwal hari : selasa  
psi | lpy3 | 07.00-09.30

===== MENU =====  
1. Tampilkan semua  
2. Cari berdasarkan hari  
3. Cari berdasarkan mata kuliah  
0. Keluar  
Pilih : 0